

Saunak Kumar Panda

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Research Interests

- **Methodologies:** Reinforcement learning; Stochastic modeling; Risk-sensitive sequential decision-making; Statistical inference and uncertainty quantification
- **Applications:** Manufacturing and industrial process control; Healthcare and treatment recommendation; Resource allocation; Procurement auctions

Education

University of Houston – PhD in Industrial Engineering, GPA: 3.96 Aug 2025

- **Specialization:** Operations Research
- **Coursework:** Deep Reinforcement Learning, Statistical Machine Learning, Stochastic Processes, Convex Optimization, Operations Research

Texas Tech University (Transferred) – PhD in Industrial Engineering, GPA: 4.0 Aug 2022

University of Washington – MS in Mechanical Engineering, GPA: 3.82 June 2020

- **Specialization:** Robotics

PES Institute of Technology – BE in Mechanical Engineering, GPA: 3.7 June 2018

- **Minor:** Electronics and Communication

Research Experience

Postdoctoral Scholar, HAI Institute, University of Houston – Houston TX Sep 2025 – *Present*

- Conduct research in reinforcement learning, machine learning, and human-centered AI, with applications in healthcare operations, recommendation systems, and algorithmic decision-making.
- Develop data-driven methods that combine predictive modeling, optimization, and domain expertise to support complex real-world decisions.

Graduate Research Assistant, University of Houston – Houston TX Aug 2020 – Aug 2025

- Conducted research with Dr. Yisha Xiang in reinforcement learning, statistical inference, and large-scale optimization for sequential decision-making.

Graduate Research Assistant, Texas Tech University – Lubbock, TX Aug 2020 – Aug 2022

BARC Graduate Researcher, Boeing Advanced Research Center – Seattle, WA Jul 2019 – Sep 2019

- Developed an automated inspection technique for tube bending using computer vision and linear regression, achieving a 75% accuracy improvement and reducing bend angle errors to ± 0.1 degrees

Precision Controls Researcher, Microsoft Research Labs – Redmond, WA Jan 2019 – Dec 2019

- Designed PID control algorithms for robotic eye-tracking systems, improving laser dot detection and calibration accuracy, and enhancing system robustness

Research Intern, Indian Institute of Technology (IIT) Bombay – Mumbai, India Jan 2017 – Jun 2017

- Achieved more efficient shape optimization of plates under various loading conditions by developing a novel variant of Particle Swarm Optimization (PSO)
- Conducted comprehensive benchmark comparisons between PSO and other contemporary optimization algorithms to evaluate performance improvements

Publications

Peer-Reviewed Publications

- [1] **Saunak Kumar Panda**, Ruiqi Liu, and Yisha Xiang, “Asymptotic Analysis of Sample-Averaged Q-Learning.”
In: *IEEE Transactions on Information Theory*, 71(7), 5601–5619, 2025.
URL: <https://doi.org/10.1109/TIT.2025.3569421>
- [2] **Saunak Kumar Panda**, Yisha Xiang, and Ruiqi Liu, “Dynamic resource matching in manufacturing using deep reinforcement learning.”
In: *European Journal of Operational Research*, 318(2), 408–423, 2024.
URL: <https://doi.org/10.1016/j.ejor.2024.05.027>
- [3] **Saunak Kumar Panda**, Ruiqi Liu, and Yisha Xiang, “Online Statistical Inference of Sample-averaged Q-Learning.”
In: *First Reinforcement Learning Safety Workshop, Reinforcement Learning Conference (RLC 2024)*.
URL: <https://openreview.net/forum?id=bKXRTxxmey>

Under Review

- [4] **Saunak Kumar Panda**, Tong Li, Yisha Xiang, and Ruiqi Liu, “Nonparametric Variance-Penalized Actor-Critic: Statistical Inference for Risk-Sensitive Reinforcement Learning.”
Submitted to *IEEE Transactions on Neural Networks and Learning Systems (IEEE TNNLS)*.
URL: <https://doi.org/10.48550/arXiv.2609.14327>
- [5] Tong Li, **Saunak Kumar Panda**, and Yisha Xiang, “Certifying Lower Bounds for Risk-Sensitive Reinforcement Learning under Adversarial State Perturbations.”
Submitted to *Machine Learning (Springer Nature)*.
URL: <https://doi.org/10.48550/arXiv.2609.10866>

Working Papers / In Preparation

- [6] **Saunak Kumar Panda**, Ying Lin, and Meng Li, “A Two-Stage Ranking Framework for ABA Treatment Goal Recommendation.”
In preparation for Journal of Biomedical Informatics (JBI).
- [7] **Saunak Kumar Panda**, Jingyi Tian, and Meng Li, “Deep RL in Procurement Auctions.”
In preparation for Manufacturing & Service Operations Management (M&SOM).
- [8] **Saunak Kumar Panda**, Masoud Heidary, Biresh Kumar Joardar, and Yisha Xiang, “Deep reinforcement learning for optimization of machine learning on manycore circuit design.”
In preparation for International Symposium on Quality Electronic Design (ISQED).
URL: <https://github.com/saunakp123/circuit-rl-optimization>

Invited & Contributed Talks

- [1] **Predictive Maintenance in Rail Operations: Railway Wheel Failure Prediction.**
 - INFORMS Annual Meeting, RAS Data ChallengeOct 2025
- [2] **Empirical Variance-Penalized Risk-Sensitive Reinforcement Learning.**
 - INFORMS Annual MeetingOct 2025
- [3] **Online Statistical Inference of time-varying sample-averaged Q-learning.**
 - INFORMS Annual MeetingOct 2024
- [4] **Online Statistical Inference of sample-averaged Q-learning.**

- Reinforcement Learning Conference (RLC) Aug 2024
- [5] **Online Statistical Inference for dynamically-changing batch Q-learning.**
 - INFORMS Annual Meeting Oct 2023
- [6] **Solving Dynamic Resource Matching in Manufacturing using Reinforcement Learning.**
 - INFORMS Annual Meeting Oct 2022
 - University of Houston Industrial Engineering Friday Seminar Series Oct 2022
- [7] **Dynamic matching of demand-supply types with manufacturing resources.**
 - Institute of Industrial and Systems Engineers Annual Conference & Expo May 2022

Grants and Proposals

National Science Foundation (NSF) – Advanced Manufacturing, Design, and Operations (AMDO)
Collaborative Research: Causal Observability and Diagnosis for Human-Robot Interactions Failures in Manufacturing

Role: Senior Personnel; **PI:** Dr. Yisha Xiang

Planned Submission: Oct 2, 2026

- Developing the task “Causal LLM reasoning-trajectory RL” to improve diagnosis of human–robot interaction failures using reinforcement learning with intermediate causal-reasoning rewards and generative training scenarios.

National Science Foundation (NSF) – Smart and Connected Health (SCH)

Collaborative Research: Smart, Secure and Scalable (S3) Health Monitoring of Large Populations via Online Collaborative Learning

Role: Other Personnel; **PI:** Dr. Ying Lin

Planned Submission: Oct 2026

- Contributing to the Human-Centered Recommendation thrust by developing interpretable treatment-goal recommendation methods for Applied Behavior Analysis, with emphasis on clinician-facing decision support and experimental evaluation.

Teaching Experience

HPE DSI 271: Data Analysis & Visualization using Excel & Power BI

Fall 2025

Instructor, Hewlett Packard Enterprise Data Science Institute, University of Houston

- Designed and taught an 11-session course (~60 students) on business and data analytics using Excel and Power BI.

INDE 7397: Decision Modeling and Optimization

Spring 2023

Curriculum Developer, Department of Industrial and Systems Engineering, University of Houston

- Developed reinforcement learning modules covering Markov decision processes, temporal-difference learning, Q-learning, and deep reinforcement learning for Dr. Yisha Xiang’s graduate course.

IE 3311: Deterministic Operations Research

Fall 2021

Teaching Assistant, Department of Industrial, Manufacturing, and Systems Engineering, Texas Tech University

- Held office hours, led problem-solving sessions in linear programming and network optimization, and graded assignments for Dr. David Huckleberry Gutman’s course.

Industry Experience

Systems Modeling & Simulation Intern, MicroVision Inc. – Redmond, WA Jan 2020 – Mar 2020

- Developed system-level models and low-level controls for automotive LiDAR systems using LTSpice, enhancing accuracy and robustness through comprehensive unit and functional testing
- Reduced model error and identified additional faults, ensuring reliable system deployment in real-world conditions

Project Intern, Robert Bosch Engineering and Business Solutions – Bangalore, India Jan 2018 – Apr 2018

- Developed an Excel tool for material quantity and price calculations, optimizing cost estimation for additive manufacturing
- Collaborated with the marketing team to design a brochure for the Additive Manufacturing division, improving communication of technical capabilities

Recognition and Awards

Finalist – INFORMS Railway Applications Section (RAS) 2025 Data Challenge Oct 2025

- Recognized among the top six international teams for developing an end-to-end predictive maintenance pipeline for railway wheel failure prediction.

Distinguished Scholarship Award – University of Houston 2023, 2024

- Recognized for research contributions and academic performance in operations research.

Service

Co-organizer, BizAI Challenge, Human-Centered AI Institute, University of Houston Nov 2025

- Co-organized an AI and business challenge for students in grades 7–12 from the Greater Houston area.

President, INFORMS Student Chapter, University of Houston Aug 2022 – Aug 2025

- Led a student chapter of over 20 members and organized seminars, research meetings, and academic events.

Student Board Member, IISE QCRE Division Aug 2022 – May 2024

- Edited and published the division's quarterly newsletter.

Conference Session Chair

- 2023 INFORMS Annual Meeting: Advances in Reinforcement Learning 2023
- IISE Annual Conference & Expo 2022: Operations Research 2022

Journal Reviewer

- European Journal of Operational Research
- Journal of Intelligent Manufacturing

Professional Affiliations

- Institute for Operations Research and the Management Sciences (INFORMS)
- Institute of Industrial and Systems Engineers (IISE)
- Institute of Electrical and Electronics Engineers (IEEE)
- INFORMS University of Houston Student Chapter